

Erdős Problem #461

EP-461

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $s_t(n)$ be the t -smooth component of n - that is, the product of all primes p (with multiplicity) dividing n such that $p < t$. Let $f(n, t)$ count the number of distinct possible values for $s_t(m)$ for $m \in [n+1, n+t]$. Is it true that $f(n, t) \gg t$ (uniformly, for all t and n)?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>